A DYNAMIC THEORY OF FAKE ϕ-FEATURES

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Puzzle: Pronouns under Focus

ϕ-features restrict the values that a pronoun can range over:

(1) Every student$_n$ gave her$_n$ mom a gift
⇒ every student is a woman

Under focus-sensitive operators (FSOs), the restriction seems to be lifted. The features are then called "fake."

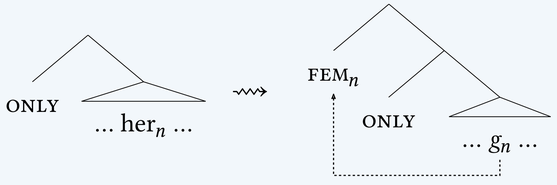
(2) Only Ann gave her mom a gift.
⇒ Bill did not give *his* mom a gift

(3) I gave my mom the cheapest gift.
⇒ you did not give *your* mom a cheaper gift

Why can *her* range over men in (2) but not in (1)? Why can *my* range over people other than the speaker in (3)?

Diagnosis

We need to evaluate ϕ-features **outside** the FSO's scope:



ϕ-Features are Dynamic Tests

MAIN CLAIMS

① A ϕ-feature is a *delayable test* on the value of a discourse referent, i.e. a post-supposition (Brasoveanu 2013).

② "Fake" features are just ϕ-features evaluated outside the scope of an FSO.

Claim ① requires passing information about binding **upwards** through the tree. We need a dynamic semantics:

- every expression accepts an input context i
- every expression outputs a set of pairs $\langle m, j\rangle$
  - m is a static meaning, j is an updated context
- Example: $[\![\text{a gift}_u]\!] = \lambda i.\{\langle g, i^{u\mapsto g}\rangle \mid \textsc{gift}\ g\}$

A ϕ-feature checks if a dref in every output context of a dynamic meaning has the right gender, number, etc.:

$$[\![\textsc{fem}_v]\!] = \lambda M\lambda i : (\forall\langle m, j\rangle \in Mi\,.\, j_v \text{ is a woman}).Mi,$$
$$[\![1\textsc{st}_v]\!] = \lambda M\lambda i : (\forall\langle m, j\rangle \in Mi\,.\, j_v = \textsc{spkr}).Mi$$

Denotations

Pronouns retrieve a dref, the operator $\triangleright_v$ stores a dref at position v in the context, and indefinites nondeterministically select an individual in their restrictor (Charlow 2014):

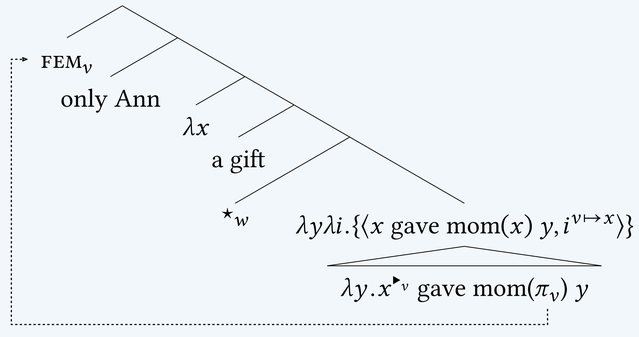
$$\textit{Pronouns}\quad [\![\pi_v]\!] = \lambda i.\{\langle i_v, i\rangle\} \qquad \textit{Indefinites}\quad [\![\text{a}]\!] = \lambda P\lambda i.\{\langle x, j\rangle \mid \langle \top, j\rangle \in Pxi\}$$
$$\textit{Binding}\quad \left[\!\left[r^{\triangleright_v}\right]\!\right] = \lambda i.\{\langle s, j^{v\mapsto s}\rangle \mid \langle s, j\rangle \in [\![r]\!]i\} \qquad \textit{Abstraction}\quad [\![\star_v]\!] = \lambda M\lambda N\lambda i.\bigcup\{Mxj^{v\mapsto x} \mid \langle x, j\rangle \in Ni\}$$

Only and *cheapest* gather all the witnesses y of their scope P, then check either that x is the only such witness, or that x is associated with the cheapest object k_v (Bumford 2017):

$$[\![\text{only}]\!] = \lambda x\lambda P\lambda i.\{\langle \top, j\rangle \mid \langle x, j\rangle \in [\![\text{a}]\!]Pi,\ \forall\langle y, k\rangle \in [\![\text{a}]\!]Pi.y = x\}$$
$$[\![\text{cheapest}_v]\!] = \lambda P\lambda x\lambda i.\{\langle \top, j\rangle \mid \langle x, j\rangle \in [\![\text{a}]\!]Pi,\ \forall\langle y, k\rangle \in [\![\text{a}]\!]Pi.\langle y, k\rangle \neq \langle x, j\rangle \Rightarrow \text{price } j_v < \text{price } k_v\}$$

Derivations

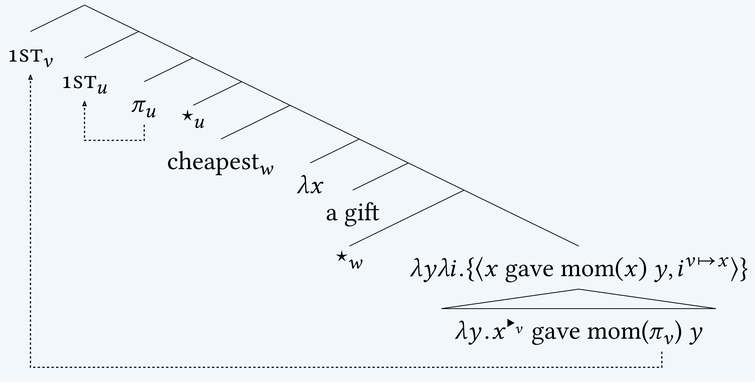
$$[\![(2)]\!] = \lambda i : \text{Ann is a woman}.\left\{\left\langle \top, i\begin{bmatrix} v\mapsto \text{Ann} \\ w\mapsto g\end{bmatrix}\right\rangle \,\middle|\, \begin{array}{c}\textsc{gift}\ g,\ \text{Ann gave mom(Ann)}\ g,\\ \forall\langle y, h\rangle.\text{gift}\ h \wedge y\ \text{gave mom}(y)\ h \Rightarrow y = \text{Ann}\end{array}\right\}$$



- *only* collects all the gift-givers, then checks that this collection contains just *Ann*
- FEM$_v$ is evaluated outside the scope of *only*

$$[\![(3)]\!] = \lambda i : i_u = \textsc{spkr}.\left\{\left\langle \top, i\begin{bmatrix} u\mapsto \textsc{spkr} \\ v\mapsto \textsc{spkr} \\ w\mapsto g\end{bmatrix}\right\rangle \,\middle|\, \begin{array}{c}\textsc{gift}\ g,\ \textsc{spkr}\ \text{gave mom}(\textsc{spkr})\ g,\\ \forall\langle y, h\rangle \neq \langle \textsc{spkr}, g\rangle.\text{gift}\ h \wedge y\ \text{gave mom}(y)\ h\\ \Rightarrow \text{price}\, g < \text{price}\, h\end{array}\right\}$$

- *cheapest* collects all pairs of gift-givers and gifts, then checks that one of SPKR's gifts is cheaper than all other gifts
- The ϕ-feature tests from *my* and from *I* are evaluated outside the scope of *cheapest*



Quantifiers

To handle (1), assume $[\![\text{every-student}_v]\!]$ is the following dynamic generalized quantifier. '#' means undefined:

$$\lambda Q\lambda i : i_v = \#.\{\langle \forall x \in \textsc{student}.\exists j.\langle \top, j\rangle \in Qxi^{v\mapsto x}, i\rangle\}$$

The post-supposition FEM$_v$ in (1) can be evaluated either **inside or outside** of the quantifier:

every-student$_v$(… FEM$_v$ …) ⇒ every student is a woman
FEM$_v$(… every-student$_v$ …) ⇒ i_v is defined and undefined

Scoping FEM$_v$ inside is the only felicitous option.

Quantificational binding of an **indexical** is impossible: #every student$_n$ decorated my$_n$ locker.

every-student$_v$(… 1ST$_v$ …) ⇒ every student is SPKR
1ST$_v$(… every-student$_v$ …) ⇒ i_v is defined and undefined

Both scopal orders are infelicitous.

Donkeys

Since it uses dynamic binding, the proposal allows for fake features in the absence of c-command (donkey sentences):

(4) Only the cop who arrested MARY insulted her.

These sentences are challenging for syntactic approaches to fake features (Kratzer 1998, Heim 2008, and others).

Summary

Earlier theories stipulated that ϕ-features can sometimes be ignored by the semantics, in one of two ways:

- ϕ-features are part of a pronoun's ordinary meaning, but not its focus meaning (Jacobson 2013)
- what looks like a pronoun is a plain variable that became syncretic with a pronoun after syntactic feature transmission (Kratzer 1998, Heim 2008)

The current proposal makes these stipulations unnecessary.

TAKEAWAYS

- Pronouns can be given a uniform meaning
- Part of that meaning is a delayable dynamic test
- Fake features under focus come about through a standard scope ambiguity